

AERO40007 – Mechanics

Imperial College of Science, Technology & Medicine

Department of Aeronautics

1. A particle at position (3, 4, 6) at time $t_0 = 1$ s is given a constant acceleration having the value $2\mathbf{i} + 1\mathbf{j}$ (m/s^2). If the velocity at time t_0 is $4.5\mathbf{i} + 6\mathbf{j} + 1.5\mathbf{k}$ (m/s), what is the position of the particle 20 seconds later?

ANS. $\mathbf{R} = 493\mathbf{i} + 324\mathbf{j} + 36\mathbf{k}$

$$\Delta \mathbf{S} = \mathbf{v}t + \frac{1}{2}\mathbf{a}t^2$$

$$= (4.5\mathbf{i} + 6\mathbf{j} + 1.5\mathbf{k}) 20 + \frac{1}{2}(2\mathbf{i} + 1\mathbf{j})(20)^2$$

$$= 90\mathbf{i} + 120\mathbf{j} + 30\mathbf{k} + 400\mathbf{i} + 200\mathbf{j}$$

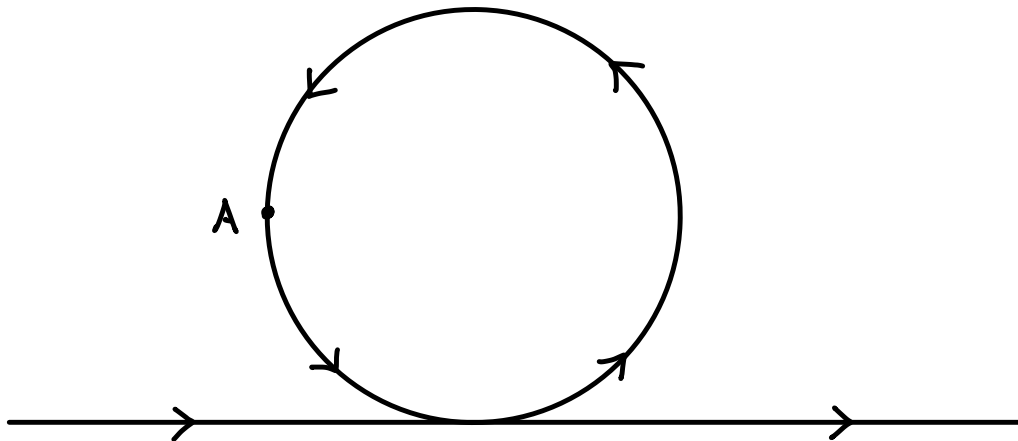
$$= 490\mathbf{i} + 320\mathbf{j} + 30\mathbf{k}$$

$$\therefore \mathbf{R} = \mathbf{S} + \Delta \mathbf{S}$$

$$= 493\mathbf{i} + 324\mathbf{j} + 36\mathbf{k}$$

2. An aeroplane is diving vertically at a speed of 900 km/h. If the maximum positive acceleration that the pilot can endure before blackout is $4g$'s, what is the minimum safe radius of curvature as he pulls out of the dive?

ANS. 2.12km



At A, aeroplane experienced the maximum acceleration

$$\therefore a_{\max} = a_{\text{circle}} + g = 4g \text{ (m/s}^2\text{)}$$

$$\therefore a_c = 4g - g = 3g \text{ (m/s}^2\text{)}$$

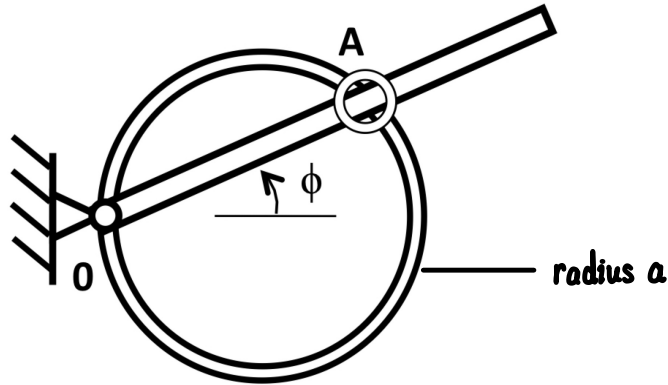
$$\therefore a_c = \frac{v^2}{r} = \frac{[(900 / 3.6) \text{ m/s}]^2}{r} = 3g \text{ m/s}^2$$

$$\therefore r = 2123.68 \text{ m}$$

$$= 2.12 \text{ km}$$

3. A small ring moves on a circular hoop of radius r . A rod OA passes through the ring and rotates about a fixed point O with constant angular velocity $\dot{\phi}$. Find the absolute acceleration of the ring.

ANS. $4a\dot{\phi}^2$



$$\dot{\phi} = \text{constant}$$

$$\therefore \mathbf{r} = R\bar{\mathbf{e}}_R + z\bar{\mathbf{e}}_z = R\bar{\mathbf{e}}_R$$

$$\therefore \frac{d\mathbf{r}}{dt} = \dot{R}\bar{\mathbf{e}}_R + R\dot{\phi}\bar{\mathbf{e}}_\phi + \dot{z}\bar{\mathbf{e}}_z = \dot{R}\bar{\mathbf{e}}_R + R\dot{\phi}\bar{\mathbf{e}}_\phi$$

$$\therefore \frac{d^2\mathbf{r}}{dt^2} = (\ddot{R} - R\dot{\phi}^2)\bar{\mathbf{e}}_R + (R\ddot{\phi} + 2\dot{R}\dot{\phi})\bar{\mathbf{e}}_\phi = (\ddot{R} - R\dot{\phi}^2)\bar{\mathbf{e}}_R + 2\dot{R}\dot{\phi}\bar{\mathbf{e}}_\phi$$

$$R = OA = 2a \cos \phi$$

$$\therefore \dot{R} = -2a\dot{\phi} \sin \phi$$

$$\ddot{R} = -2a\dot{\phi}^2 \cos \phi$$

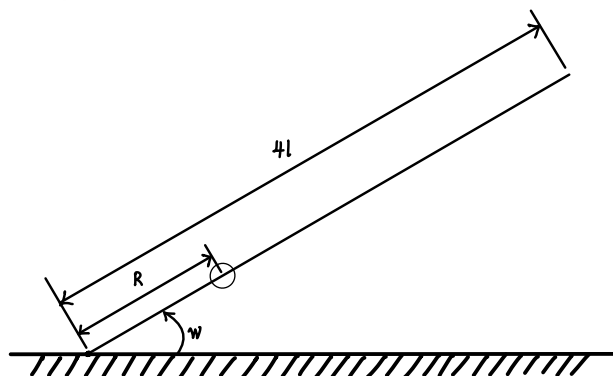
$$\therefore \frac{d^2\mathbf{r}}{dt^2} = (-2a\dot{\phi}^2 (\cos \phi - 2\cos \phi))\bar{\mathbf{e}}_R - 4a\dot{\phi}^2 \sin \phi \bar{\mathbf{e}}_\phi$$

$$= (4a\dot{\phi}^2)(-\cos \phi \bar{\mathbf{e}}_R - \sin \phi \bar{\mathbf{e}}_\phi)$$

$$\therefore |\mathbf{a}| = \sqrt{\left(\frac{d^2\mathbf{r}}{dt^2}\right)^2} = 4a\dot{\phi}^2 (\sqrt{\cos^2 \phi + \sin^2 \phi})$$

$$= 4a\dot{\phi}^2$$

4. A smooth rod of length $4l$ rotates at a constant angular velocity ω , in a horizontal plane, about a pivot at one end. A smooth ring of mass m is free to slide along the rod. It is initially fixed a distance l from the pivot. At time $t = 0$ the ring is released. Obtain the equation of motion of the ring and show that the ring leaves the rod after a rotation of approximately 120 degrees. What is the power required to drive the rod as a function of the time.



$$\mathbf{r} = R\mathbf{\bar{e}}_R \quad \therefore \dot{\mathbf{r}} = \dot{R}\mathbf{\bar{e}}_R + R\dot{\phi}\mathbf{\bar{e}}_\phi$$

$$\begin{aligned} \therefore \ddot{\mathbf{r}} &= \ddot{R}\mathbf{\bar{e}}_R + \dot{R}\dot{\phi}\mathbf{\bar{e}}_\phi + \dot{R}\dot{\phi}\mathbf{\bar{e}}_\phi + R\ddot{\phi}\mathbf{\bar{e}}_\phi + (-R\dot{\phi}^2\mathbf{\bar{e}}_R) \\ &= (\ddot{R} - R\dot{\phi}^2)\mathbf{\bar{e}}_R + (2\dot{R}\dot{\phi} + R\ddot{\phi})\mathbf{\bar{e}}_\phi \end{aligned}$$

$$\therefore \dot{\phi} = \omega \quad \therefore \ddot{\phi} = \text{constant} \quad \therefore \ddot{\mathbf{r}} = (\ddot{R} - R\dot{\phi}^2)\mathbf{\bar{e}}_R + (2\dot{R}\dot{\phi})\mathbf{\bar{e}}_\phi$$

$$F = \frac{f}{m} \quad \therefore \text{No force acting along } \mathbf{\bar{e}}_R \quad \therefore \ddot{R} - R\dot{\phi}^2 = 0$$

$$\text{For } a\frac{dy}{dt} + b\frac{dy}{dt} + cy = 0, \quad y = Ae^{mt} + Be^{nt}$$

$$\text{Let } R = e^{mt} \quad \therefore m^2e^{mt} - \omega^2e^{mt} = 0 \quad \therefore e^{mt}(m^2 - \omega^2) = 0$$

$$\therefore e^{mt} > 0 \quad \therefore m = \pm\omega \quad \therefore R = Ae^{\omega t} + Be^{-\omega t} \quad \therefore R' = A\omega e^{\omega t} - B\omega e^{-\omega t}$$

$$\text{When } t=0, \quad R = A + B = l, \quad R' = (A - B)\omega = 0 \quad \therefore A - B = 0$$

$$\therefore A = B = \frac{l}{2} \quad \therefore R = \frac{l}{2}(e^{\omega t} + e^{-\omega t}) = l \cosh(\omega t)$$

$$\text{When } R = 4l, \quad \cosh(\omega t) = 4 \quad \therefore e^{\omega t} + \frac{1}{e^{\omega t}} = 8 \quad \therefore e^{\omega t} = 4 \pm \sqrt{15}$$

$$\therefore \omega t > 0 \quad \therefore e^{\omega t} = 4 + \sqrt{15} \quad \therefore \omega t = \theta = \ln(4 + \sqrt{15}) = 116.23^\circ = 120^\circ$$

$$\therefore F = m\mathbf{a} = m\ddot{\mathbf{r}} \quad \therefore \dot{R} = \frac{l}{2}(\omega e^{\omega t} - \omega e^{-\omega t}) = \frac{l\omega}{2}(e^{\omega t} - e^{-\omega t})$$

$$\therefore F = m\omega l(e^{\omega t} - e^{-\omega t})\mathbf{\bar{e}}_\phi = m\omega l(e^{\omega t} - e^{-\omega t})$$

$$\therefore P = Fv = F\omega r = m\omega l(e^{\omega t} - e^{-\omega t}) \cdot R \cdot \omega = m\omega^2 l^2 \sinh(2\omega t)$$